Sharpness: Check the image for sharpness and detail. The will be particularly apparent in images with a lot of texture (cloth or a painted wall, dirt, sand, etc.) or foliage.

Banding: Smooth colour transition is a must for a colour printer and this is where the black to white transition strip in the test image mentioned earlier comes in. That strip needs to be perfectly smooth with no visible colour bands visible.

Tonal accuracy: This is a little harder to judge, but if you're using an image that showcases a colour tone that you're intimately familiar with (skin tones, the colour of your favourite shirt, etc.), you'll get some idea of the printer's capabilities.

Ink bleed: This happens when the printer is pushing out too much ink onto the paper. Of course, you need to ensure that you've selected the right kind of paper in the printer's options menu, but after that, there shouldn't be any noticeable ink-bleed. This is most prominent in text rather than in images, so take a test print with text in various sizes and colours and then use a loupe to check for bleed.

2. Laser printers

Since these printers are mostly for office use and laser printers largely offer the exact same print quality, you're only really looking for reliability and suitability to your work environment, not to mention operating costs.

Print volume: You need to determine what your monthly print volume is as all laser printers are rated based on that. Some will have a recommended volume of 500 pages per month, others, more. If you're going to be printing a lot of content, just grab a printer rated for high-volume as these are designed to handle that extra wear and tear.

Printing costs: This is an aspect that we'll touch on a little later, but suffice to say that when printing a lot, cost of operation is an important factor that will determine your purchase. Do check if your manufacturer offers high-yield cartridges or some sort of official refill option as this will considerably cut down on your printing costs.

Print speed: This will not matter if you're printing barely 10-20 pages at a time, but if you regularly print larger documents, then you'll need to check this aspect. Take a heavy document (with lots of text and some images) and set the printer to print a 100 pages (don't worry, this isn't too expensive and that much paper will only cost ₹50 or less). Printers with a low quantity of RAM will struggle after about 20 pages and you'll see these printers pause to take a breather. If that doesn't bother you, you can ignore it, if it does, take a printer with more RAM.

Startup time: You may not realise this, but many laser printers take quite a bit of time to startup and the first page may not be out for even 30 seconds. If this is something that will bother you, turn on a printer, let it go into its sleep state and then give it a print command. The time to the first print is what you're looking at.

Print quality: As mentioned earlier, this shouldn't be your primary reason for purchasing a laser printer, but, there is some variation in print quality and if you're really finicky about it, create a document with text sizes ranging from 1pt to 40pts in a font that you know you'll be using. Also, don't forget to include the point size that you'll be using regularly. Take a print of that document from various printers and use a magnifying lens or loupe to check the quality of the print. You're looking for overlapping characters and ink-bleed.

3. Ink-tank printers

These printers differ from inkjet and laser printers in that they squirt liquid ink onto the page. This ink is cheap, so the cost of printing is also cheap, but these printers can be expensive, making them more suited to high volume prints. All the tests mentioned previously apply to these printers as well with one exception. These printers are NOT meant for photo prints on any sort of paper. These printers are just meant to give cheap, colour prints.

The most important thing you need to keep a lookout for here is ink-bleed as this is a serious problem with this tech. When going for this printer, it's always a trade-off between ink-bleed and cost and usually, cost wins.

Cost of ownership

We've left this out for last, but particularly when it comes to ink-tank and laser printers, this is the most important aspect to consider. Most manufacturers claim that their print rates are around ₹1-5 per page, and that's true, but that rate is almost exclusively determined by the cost of the cartridge and its yield. What do you do when you have a choice between a ₹5,000 printer that yields prints at ₹1 per page and one that costs ₹10,000 and yields prints at ₹0.5 per page?

Here's what you do (for a laser printer):

- Cost of printer: X
- Cost of cartridge: Y
- Page yield of cartridge: Z
- Your estimated print volume per year: A
- Estimated life of printer (years): 3 (going by the average warranty period, but you can modify this as required.)
- Estimated print volume over the life of the printer: 3*A
- Number of cartridges required over the life cycle of the printer: (3*A)/Z
- Total cost of cartridges required: B=Y*((3*A)/Z)

Therefore, the Cost of Ownership would be as follows:

COO = (X + B)/(3*A)

Everything else being equal, the lower this value, the better. For colour printers, you need to estimate the number of colour vs. black and white prints that you'd use and expand the formula to account for the cost of colour cartridges. Especially if you'll need CMYK cartridges.

How DO we test printers?

All of the above, plus a detailed record of printing time under a variety of load conditions. More importantly, we use a Datacolour SpyderPRINT (http://dgit.in/1d9px4V) unit worth ₹25,000 to create an ICC profile for a printer and verify colour accuracy.